Fatima Jinnah Women University



Department of Computer Science
DevOps Lab # 7

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Task 1: Install Docker and Docker Compose

```
hina-ali32@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
hina-ali32@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
hina-ali32@ubuntu:~$
```

Shows the Docker and Docker Compose version checks with both versions displayed in the Ubuntu terminal.

Task 2: Run Two PostgreSQL Versions in Docker

```
hina-ali32@ubuntu:~$ docker pull postgres:9.6
9.6: Pulling from library/postgres
4b9a74ac6ea0: Pull complete
01c676df543a: Pull complete
1cb79db8a9e7: Pull complete
2d87c3a4038c: Pull complete
1e3d335ef0b7: Pull complete
f955a6cf127b: Pull complete
f6bae7873dd7: Pull complete
f62dc55c568d: Pull complete
8f7722dc50a7: Pull complete
e8622b8cb6f3: Pull complete
d6d74bba3a57: Pull complete
4e2c4902efbd: Pull complete
874d4d2a09fd: Pull complete
11087f2b0d87: Pull complete
Digest: sha256:caddd35b05cdd56c614ab1f674e63be778e0abdf54e71a7507ff3e28d4902698
Status: Downloaded newer image for postgres:9.6
docker.io/library/postgres:9.6
hina-ali32@ubuntu:~$ docker run postgres:9.6
Error: Database is uninitialized and superuser password is not specified.
       You must specify POSTGRES_PASSWORD to a non-empty value for the
       superuser. For example, "-e POSTGRES_PASSWORD=password" on "docker run".

       You may also use "POSTGRES_HOST_AUTH_METHOD=trust" to allow all
       connections without a password. This is *not* recommended.

       See PostgreSQL documentation about "trust":
       https://www.postgresql.org/docs/current/auth-trust.html
hina-ali32@ubuntu:~$
```

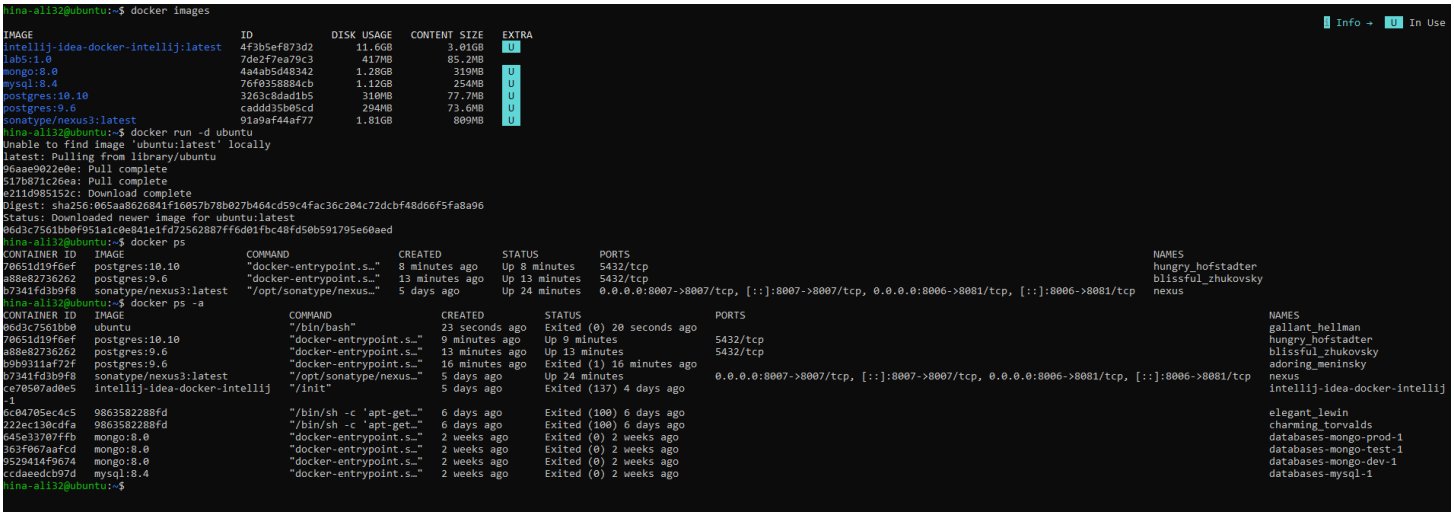
Shows docker run postgres:9.6 exiting with the missing-password error.

```
hina-ali32@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:9.6
a88e82736262afaf2e4ca47001101adb71cd0a93a33e4d0ca969aca8e3b8b301
hina-ali32@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS          PORTS                                                
```

Shows the docker info output with the Docker server details.

```
hina-ali32@ubuntu:~$ docker search ubuntu
NAME                      DESCRIPTION                                     STARS     OFFICIAL
ubuntu                    Ubuntu is a Debian-based Linux operating sys…   17878     [OK]
ubuntu/squid              Squid is a caching proxy for the Web. Long-t…   132
ubuntu/nginx              Nginx, a high-performance reverse proxy & we…   143
ubuntu/bind9              BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka              Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper          ZooKeeper maintains configuration informatio…   15
ubuntu/apache2            Apache, a secure & extensible open-source HT…   106
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   34
ubuntu/prometheus         Prometheus is a systems and service monitori…   80
ubuntu/jre                Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet      Chiselled Ubuntu runtime image for ASP.NET a…   28
ubuntu/mysql              MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-runtime     Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/postgres           PostgreSQL is an open source object-relation…   46
ubuntu/dotnet-deps        Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/redis              Redis, an open source key-value store. Long-…   26
ubuntu/jdk                Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq           Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/memcached          Memcached, in-memory keyvalue store for smal…   8
ubuntu/grafana            Grafana, a feature rich metrics dashboard & …   13
ubuntu/loki               Grafana Loki, a log aggregation system like …   3
ubuntu/traefik            Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/mimir              Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0
ubuntu/php                PHP is a general-purpose scripting language …   10
hina-ali32@ubuntu:~$
```

Shows the docker search ubuntu results with the official ubuntu image.



```
hina-ali32@ubuntu:~$ docker images
                                                
```

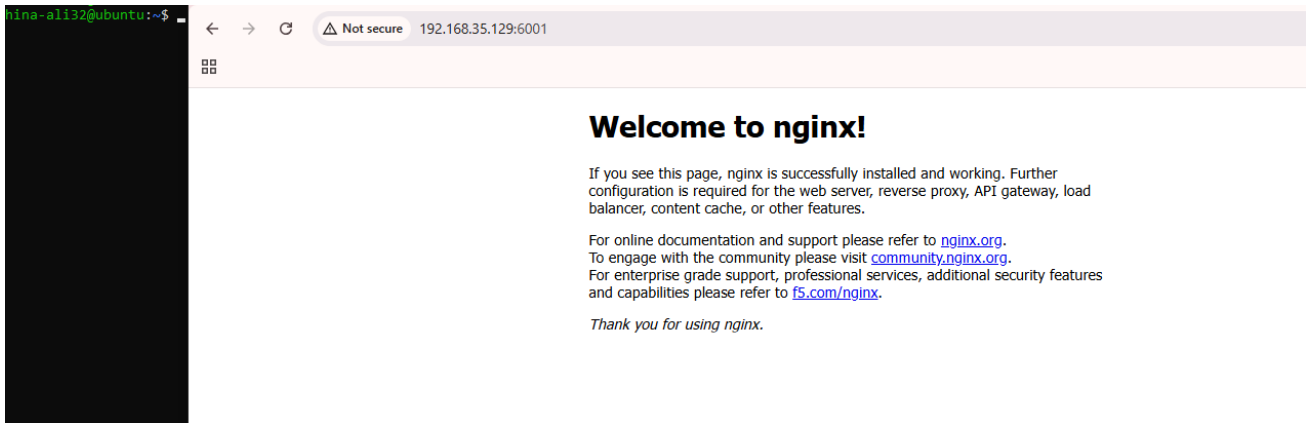
```
hina-ali32@ubuntu:~$ docker rm adoring_meninsky
adoring_meninsky
hina-ali32@ubuntu:~$ docker restart blissful_zhukovsky
blissful_zhukovsky
hina-ali32@ubuntu:~$ docker stop  blissful_zhukovsky
blissful_zhukovsky
hina-ali32@ubuntu:~$ docker rm blissful_zhukovsky
blissful_zhukovsky
hina-ali32@ubuntu:~$ docker rmi postgres:9.6
Untagged: postgres:9.6
Deleted: sha256:caddd35b05cdd56c614ab1f674e63be778e0abdf54e71a7507ff3e28d4902698
hina-ali32@ubuntu:~$
```

Shows the postgres:9.6 container restarted, stopped and removed, followed by removal of its image.

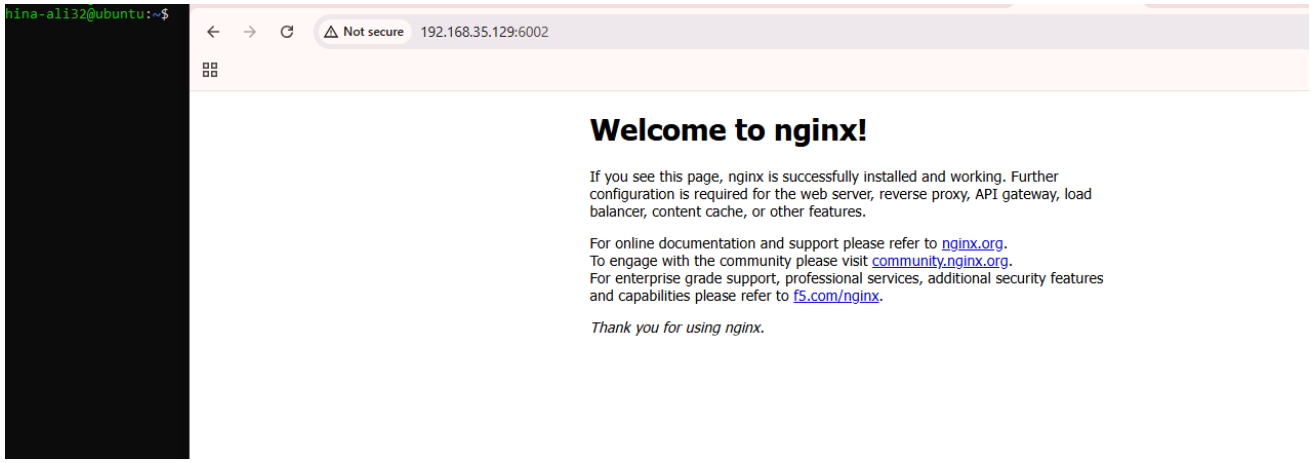
Task 4: Run Two NGINX Containers on Different Ports

```
hina-ali32@ubuntu:~$ docker run -dit -p 6001:80 --name nginx-latest nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
6d2e43921e9e: Pull complete
1724c05f72c6: Pull complete
37fb86c2a767: Pull complete
512ac5e7ce8e: Pull complete
1962e1ff75b4: Pull complete
7cb4701ce7cd: Pull complete
bbb080c71d50: Pull complete
4582dace95e2: Download complete
d2c7e31f04a3: Download complete
Digest: sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82
Status: Downloaded newer image for nginx:latest
faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e
hina-ali32@ubuntu:~$ docker run -d -p 6002:80 --name nginx-older nginx:1.30.6
Unable to find image 'nginx:1.30.6' locally
1.30.6: Pulling from library/nginx
2fc2bc44a8d3: Pull complete
cbbdc282ecf5: Pull complete
b532f9701850: Pull complete
f7214361b33f: Pull complete
9c0452afd4ee: Pull complete
89e2dd8a6236: Pull complete
1e24db3ea44b: Download complete
47835a5e6782: Download complete
Digest: sha256:42c68eab6f7e22611786a9fe1d1c019e275c50185bd54b8f75dc2b9184322703
Status: Downloaded newer image for nginx:1.30.6
b82d7c9141130bbc8e65787d234b8fed2b64e07e0b1ca546de54420ad8af17ea
hina-ali32@ubuntu:~$
```

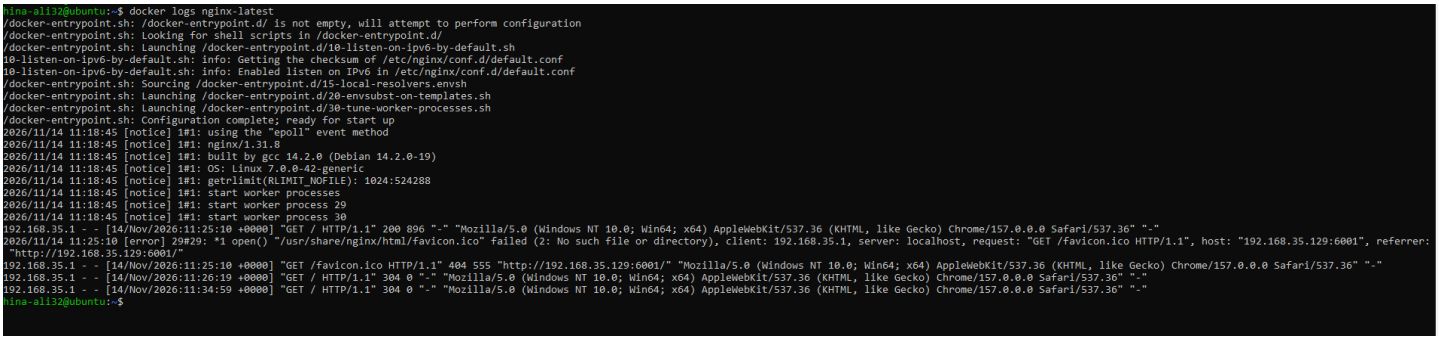
Shows both NGINX containers started on ports 6001 and 6002 with their container IDs.



Shows the default NGINX welcome page served by nginx-latest on port 6001.

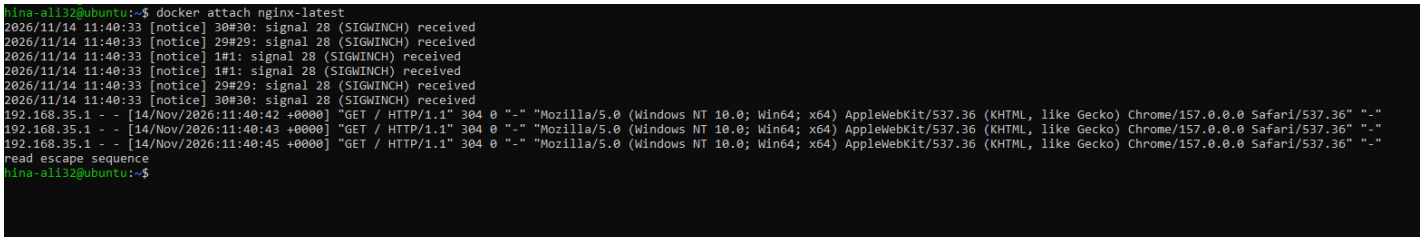


Shows the default NGINX welcome page served by nginx-older on port 6002.

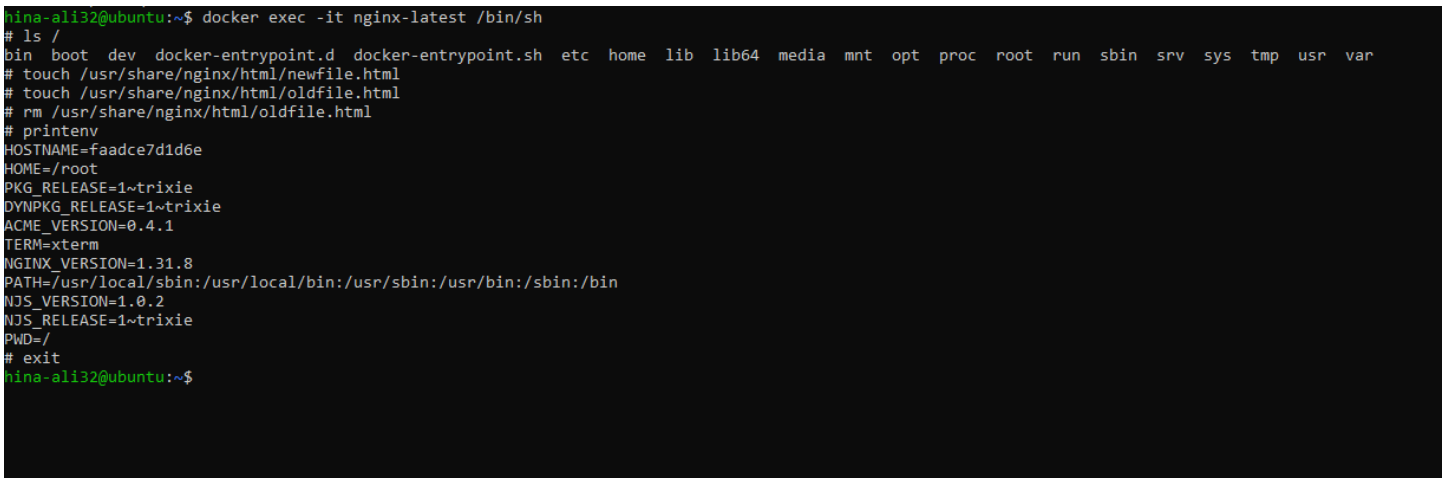


Shows the nginx-latest logs with the access log entry for the browser request.

Task 5: Work Inside a Container and Save a New Image



Shows attaching to nginx-latest, the new request logs, and detaching back to the Ubuntu prompt.



Shows a shell opened inside nginx-latest with files created, removed and the environment variables listed.

```
hina-ali32@ubuntu:~$ docker diff nginx-latest
C /run
A /run/nginx.pid
C /usr
C /usr/share
C /usr/share/nginx
C /usr/share/nginx/html
A /usr/share/nginx/html/newfile.html
C /etc
C /etc/nginx
C /etc/nginx/conf.d
C /etc/nginx/conf.d/default.conf
C /var
C /var/cache
C /var/cache/nginx
A /var/cache/nginx/client_temp
A /var/cache/nginx/fastcgi_temp
A /var/cache/nginx/proxy_temp
A /var/cache/nginx/scgi_temp
A /var/cache/nginx/uwsgi_temp
hina-ali32@ubuntu:~$
```

Shows docker diff listing newfile.html as an added file.


```
hina-ali32@ubuntu:~$ docker inspect nginx-latest
[
    {
        "Id": "faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e",
        "Created": "2026-11-14T11:18:44.66505539Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 4433,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-11-14T11:18:45.113162427Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82",
        "ResolvConfPath": "/var/lib/docker/containers/faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e/hostname",
        "HostsPath": "/var/lib/docker/containers/faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e/hosts",
        "LogPath": "/var/lib/docker/containers/faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e/faadce7d1d6e4d4b300070c10a79292377233464491f123046d1fb247385d39e-json.log",
        "Name": "/nginx-latest",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "docker-default",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
```


Shows the docker inspect output with the container state as running.

```
hina-ali32@ubuntu:~$ docker commit nginx-latest nginx:myimage
sha256:5b51475a8fd3fdc95972894356af1f77f9e92084009a59a5f6909fe91d6ce4d0
hina-ali32@ubuntu:~$
```

Shows the nginx-latest container committed as the new image nginx:myimage with its image ID.